

```
!nvidia-smi
```

```
Fri Feb 27 19:34:57 2026
```

```
+-----+
| NVIDIA-SMI 580.82.07                Driver Version: 580.82.07          CUDA Version: 12.0
+-----+-----+-----+-----+-----+
| GPU   Name                               Persistence-M | Bus-Id        Disp.A | Volatile Un- |
| Fan   Temp   Perf              Pwr:Usage/Cap |               Memory-Usage | GPU-Util  C |
|=====+=====+=====+=====+=====+
|    0  Tesla T4                       Off          | 00000000:00:04.0 Off  |             |
| N/A   44C    P8              10W /  70W      |  0MiB / 15360MiB      |           0% |
+-----+-----+-----+-----+-----+

+-----+
| Processes:
| GPU   GI    CI          PID    Type   Process name                      G
|      ID    ID                                   |                          U
+-----+
|  No running processes found
+-----+
```

```
# Après avoir monté le drive
import zipfile
import os

# Vérifier que le fichier existe
zip_path = '/content/drive/MyDrive/matricule.zip'
if os.path.exists(zip_path):
    # Extraire le zip
    with zipfile.ZipFile(zip_path, 'r') as zip_ref:
        zip_ref.extractall('/content/matricule')
    print("Fichier extrait avec succès dans /content/matricule")
else:
    print(f"Le fichier {zip_path} n'existe pas")
```

```
Le fichier /content/drive/MyDrive/matricule.zip n'existe pas
```

```
# Méthode 2: Lire avec Python
import yaml

with open('/content/matricule/data.yaml', 'r') as f:
    data = yaml.safe_load(f)
    print("Contenu de data.yaml:")
    print(data)

# Afficher de façon plus lisible
print("\nFormat YAML original:")
!cat /content/matricule/data.yaml
```

```
Contenu de data.yaml:  
{'train': '../train/images', 'val': '../valid/images', 'test': '../test/images',  
  
Format YAML original:  
train: ../train/images  
val: ../valid/images  
test: ../test/images  
  
nc: 13  
names: ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'RS', 'TU', 'libye']  
  
roboflow:  
  workspace: sahnouns-workspace  
  project: matricule-xfgoq-mjwv0  
  version: 1  
  license: CC BY 4.0  
  url: https://universe.roboflow.com/sahnouns-workspace/matricule-xfgoq-mjwv0/dataset
```

```
from google.colab import drive  
drive.mount('/content/drive')  
!ls "/content/drive/MyDrive/"
```

```

part2.mp4
pdf.pdf
'Plan_Etude(GLID).pdf'
'Présentationpfe Corrigée (1).pptx.gslides'
'Présentationpfe Corrigée.pptx.gslides'
prog_python_ex2tp3.pdf
projet-arch-1-1.gdoc
'projet data_mining.zip'
projetMb
projetMb.zip
rang.pdf
Rapport_E-Sante.gdoc
'Sahnou Mahdi - Compte rendu TD6.gdoc'
Scan-14-Aug-24-080005.pdf
Scan-14-Aug-24-081324.pdf
Screenshot_2020-06-26-23-12-34.png
Screenshot_2020-07-08-19-59-59.png
Screenshot_2020-07-19-14-13-00.png
Screenshot_2020-07-30-07-31-44.png
Screenshot_2020-07-31-17-48-50.png
Screenshot_20200809-214107_Lite.jpg
sys-exploitation
'template examen tp.gdoc'
tp2cloud.pptx
tp4.zip
'tp5 java.docx'
'Tp N 11 - 2022.gdoc'
'Untitled document.gdoc'
'Untitled presentation.gslides'

```

```
!pip install ultralytics
```

```

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dis
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
  Downloading ultralytics_thop-2.0.18-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/pyt

```

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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/pyt
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/di
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/l
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/l
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Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/l
Requirement already satisfied: nvidia-cuspars-cu12==12.5.8.93 in /usr/local/l
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/pyth
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/py
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/pyt
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/li
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: cuda-pathfinder~1.1 in /usr/local/lib/python3.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/di
Downloading ultralytics-8.4.18-py3-none-any.whl (1.2 MB)
1.2/1.2 MB 32.0 MB/s eta 0:00:00
Downloading ultralytics_thop-2.0.18-py3-none-any.whl (28 kB)
Installing collected packages: ultralytics-thop, ultralytics
Successfully installed ultralytics-8.4.18 ultralytics-thop-2.0.18

```

```
!yolo detect train data=/content/matricule/data.yaml model=yolo11s.pt epochs=60
```

58/60	4.9G	0.6686	0.3058	0.891	45	640:
	Class	Images	Instances	Box(P	R	mAP50
	all	154	1173	0.975	0.985	0.985

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
59/60	4.9G	0.6664	0.3059	0.8945	42	640:
	Class	Images	Instances	Box(P	R	mAP50
	all	154	1173	0.975	0.984	0.984

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
60/60	4.9G	0.6592	0.3056	0.8918	45	640:
	Class	Images	Instances	Box(P	R	mAP50
	all	154	1173	0.974	0.984	0.984

60 epochs completed in 0.228 hours.

Optimizer stripped from /content/runs/detect/train/weights/last.pt, 19.2MB

Optimizer stripped from /content/runs/detect/train/weights/best.pt, 19.2MB

Validating /content/runs/detect/train/weights/best.pt...

Ultralytics 8.4.18  Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 1491

YOLO11s summary (fused): 101 layers, 9,417,831 parameters, 0 gradients, 21.3 G

	Class	Images	Instances	Box(P	R	mAP50
	all	154	1173	0.974	0.984	0.984
	0	65	75	0.999	0.987	0.995
	1	125	196	0.995	0.972	0.995
	2	75	101	0.99	0.99	0.995
	3	81	102	1	0.933	0.99
	4	69	95	0.999	1	0.995
	5	76	90	0.971	1	0.995
	6	68	91	0.989	0.982	0.994
	7	69	94	1	0.989	0.995
	8	64	85	0.998	0.988	0.994
	9	72	91	0.989	0.971	0.984
	RS	1	1	0.894	1	0.995
	TU	146	146	1	0.978	0.995
	libye	6	6	0.844	1	0.876

Speed: 0.2ms preprocess, 5.6ms inference, 0.0ms loss, 5.1ms postprocess per im

Results saved to **/content/runs/detect/train**

 Learn more at <https://docs.ultralytics.com/modes/train>

Double-click (or enter) to edit

```
!yolo detect predict model=runs/detect/train/weights/best.pt source=matricule/v
```

```

image 111/154 /content/matricule/valid/images/1-618-_auto_jpg.rf.8180c6f9d2af9
image 112/154 /content/matricule/valid/images/1-622-_auto_jpg.rf.edc5fc0764159
image 113/154 /content/matricule/valid/images/1-632-_auto_jpg.rf.b1f2144669e64
image 114/154 /content/matricule/valid/images/1-636-_auto_jpg.rf.f972896f8e7a2
image 115/154 /content/matricule/valid/images/1-641-_auto_jpg.rf.6af5ecccc4234
image 116/154 /content/matricule/valid/images/1-643-_auto_jpg.rf.636a2c96ee7fc
image 117/154 /content/matricule/valid/images/1-654-_auto_jpg.rf.63126feca2ebc
image 118/154 /content/matricule/valid/images/1-658-_auto_jpg.rf.f221679fb4946
image 119/154 /content/matricule/valid/images/1-678-_auto_jpg.rf.ab812e1643596
image 120/154 /content/matricule/valid/images/1-680-_auto_jpg.rf.96be5c8c3071f
image 121/154 /content/matricule/valid/images/1-687-_auto_jpg.rf.125f728727f09
image 122/154 /content/matricule/valid/images/1-691-_auto_jpg.rf.b3bc230349217
image 123/154 /content/matricule/valid/images/1-697-_auto_jpg.rf.92d831e4d82b8
image 124/154 /content/matricule/valid/images/1-698-_auto_jpg.rf.559889a6383bb
image 125/154 /content/matricule/valid/images/1-700-_auto_jpg.rf.ffb6b4c805ed4e
image 126/154 /content/matricule/valid/images/1-718-_auto_jpg.rf.e7cc413fbdd3f
image 127/154 /content/matricule/valid/images/1-724-_auto_jpg.rf.69f20392957de
image 128/154 /content/matricule/valid/images/1-725-_auto_jpg.rf.d07575aedf650
image 129/154 /content/matricule/valid/images/1-729-_auto_jpg.rf.4ccb529708b4e
image 130/154 /content/matricule/valid/images/1-730-_auto_jpg.rf.a3e96125eb959
image 131/154 /content/matricule/valid/images/1-731-_auto_jpg.rf.88aebd61bf42
image 132/154 /content/matricule/valid/images/1-732-_auto_jpg.rf.d1c8d09f4edde
image 133/154 /content/matricule/valid/images/1-743-_auto_jpg.rf.1a1a941c2a97d
image 134/154 /content/matricule/valid/images/1-744-_auto_jpg.rf.fff7247cecb3a
image 135/154 /content/matricule/valid/images/1-760-_auto_jpg.rf.fc9c990bc861f
image 136/154 /content/matricule/valid/images/1-766-_auto_jpg.rf.281ca87e61cd7
image 137/154 /content/matricule/valid/images/20230321172016_plate_jpg.rf.97c6
image 138/154 /content/matricule/valid/images/20230321172032_plate_jpg.rf.aa4e
image 139/154 /content/matricule/valid/images/20230321172036_plate_jpg.rf.0f71
image 140/154 /content/matricule/valid/images/20230321172217_plate_jpg.rf.1d67
image 141/154 /content/matricule/valid/images/20230321172246_plate_jpg.rf.0aa7
image 142/154 /content/matricule/valid/images/20230321172528_plate_jpg.rf.2458
image 143/154 /content/matricule/valid/images/20230321172803_plate_jpg.rf.9565
image 144/154 /content/matricule/valid/images/4RLA1T7PDW_auto_jpg.rf.00a729ad4
image 145/154 /content/matricule/valid/images/64KREIU8RB_auto_jpg.rf.055d52148
image 146/154 /content/matricule/valid/images/D3T24Z2X7D_auto_jpg.rf.ec37611ba
image 147/154 /content/matricule/valid/images/F26CA04SSZ_auto_jpg.rf.2ea20d647
image 148/154 /content/matricule/valid/images/GET10IEABH_auto_jpg.rf.ac0fd8403
image 149/154 /content/matricule/valid/images/MCHTTS0846_auto_jpg.rf.a7914da44
image 150/154 /content/matricule/valid/images/R7QPW2BN3B_auto_jpg.rf.c785fef6d
image 151/154 /content/matricule/valid/images/UE3530K5AG_auto_jpg.rf.60c927075
image 152/154 /content/matricule/valid/images/UYNQ582WWG_auto_jpg.rf.fe924ccc2
image 153/154 /content/matricule/valid/images/VYJG4PTFTJ_auto_jpg.rf.7157a993e
image 154/154 /content/matricule/valid/images/YZNJKIV5Z3_auto_jpg.rf.455758a35
Speed: 1.3ms preprocess, 8.5ms inference, 1.2ms postprocess per image at shape
Results saved to /content/runs/detect/predict

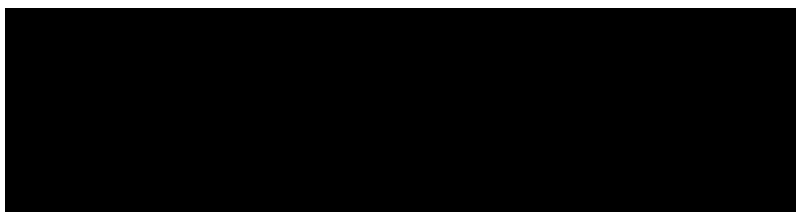
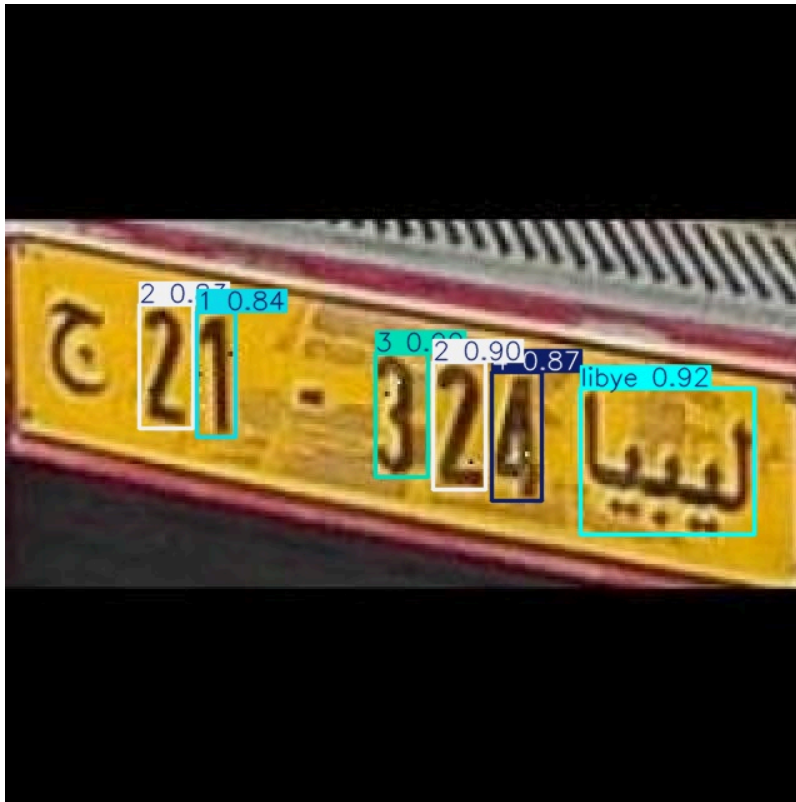
```

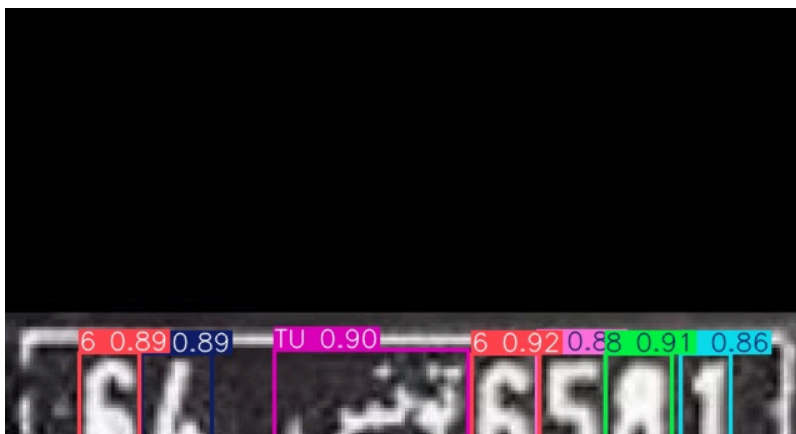
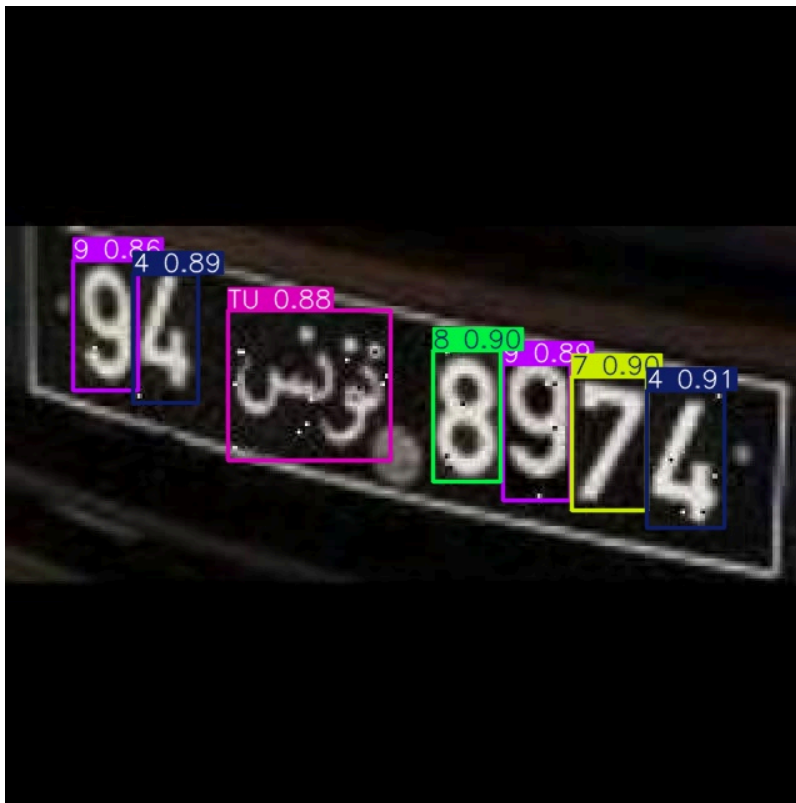
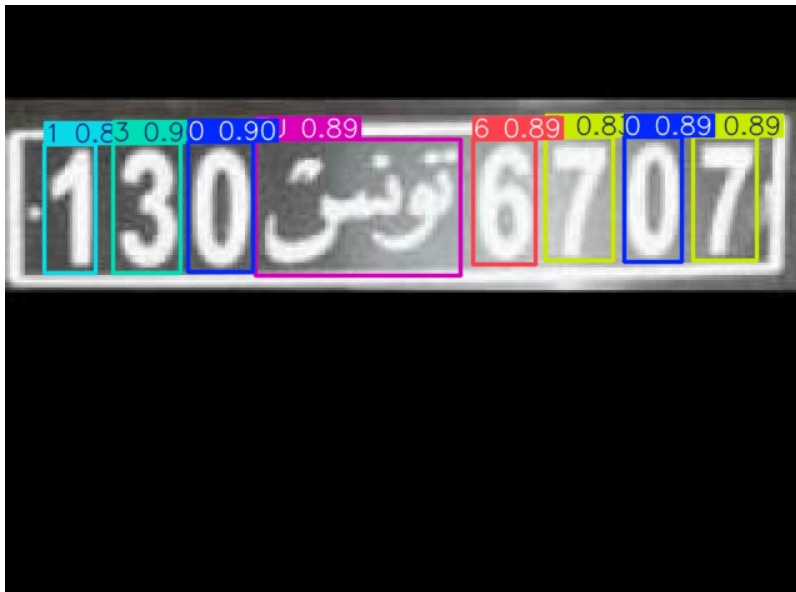
💡 Learn more at <https://docs.ultralytics.com/modes/predict>

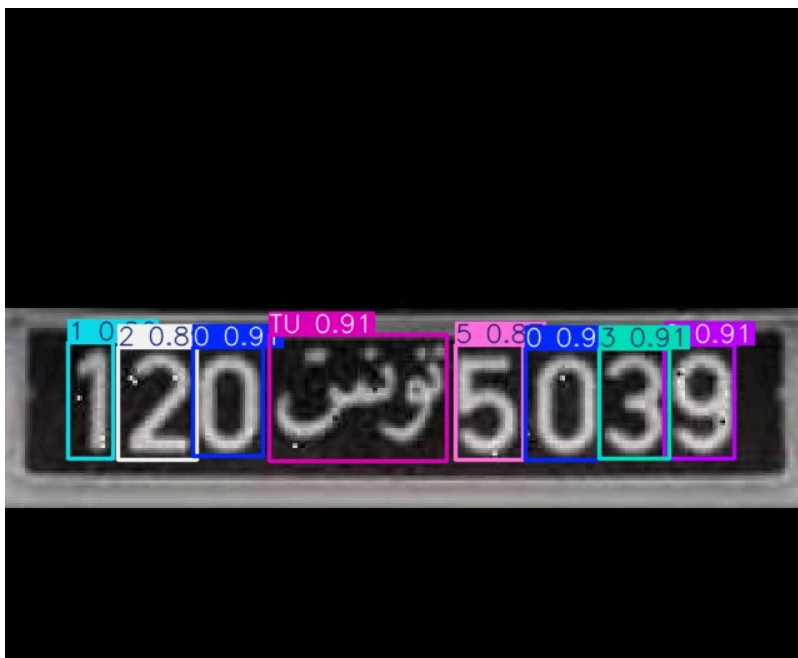
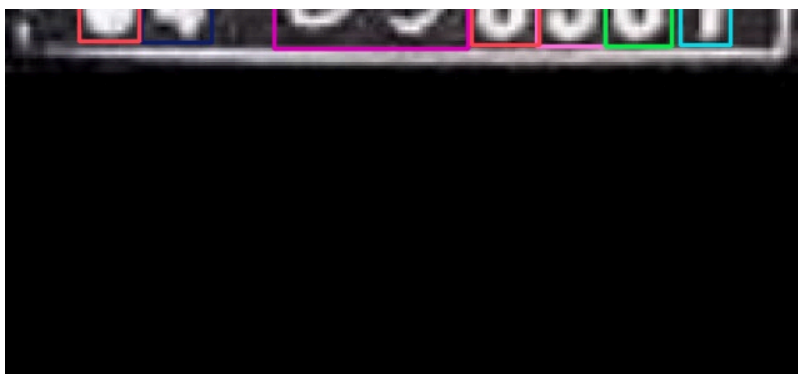
```

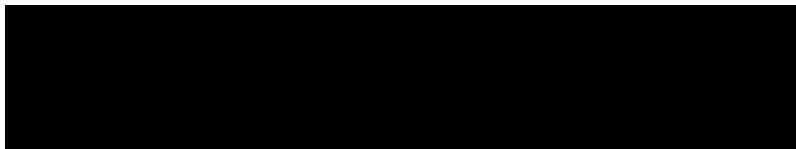
import glob
from IPython.display import Image, display
for image_path in glob.glob(f'/content/runs/detect/predict/*.jpg')[:10]:
    display(Image(filename=image_path, height=400))
    print('\n')

```







Double-click (or enter) to edit

